

A Growing-Cutoff and Convex-Profile Transfer Audit for the V59 Residual

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Abstract

TPC-268 showed that a fixed finite comparison cutoff can move a literal V59 residual across the quarter threshold. We now replace that fixed choice by the registered finite proxy $z_N = \lfloor \log N \rfloor$ and move the kernel along the convex family $K_\theta = (1 - \theta)K_{H,1} + \theta K_{H,2}$ generated by two normalized nonnegative profiles. The physical prime shell, masks, deleted diagonal, source coefficient, and rank-three projection remain unchanged. Outward rational intervals classify twelve rows: eight contractions and four obstructions. At the same growing-cutoff clock, the central profile path has $\theta = 9/10$ above and $\theta = 24/25$ below $1/4$. This is a finite model-relative transfer obstruction, not an asymptotic V59 estimate.

1 Position and claim firewall

The residual object left by the typed TPC-266 chain is a literal prime-shell quantity whose radius and phase must be controlled at growing scale. TPC-267 instantiated that object at twelve finite rows with a $z = 2$ comparison and found a strict quarter contraction. TPC-268 then demonstrated that changing declared finite cutoff, clock, or kernel parameters can reverse the verdict. The present paper asks the next natural question: does a declared scale-following cutoff together with a convex profile path remove that finite sensitivity?

We use the following status label:

NUMERICALLY_CERTIFIED_FINITE_GROWING_CUTOFF_PROFILE_TRANSFER

The word “growing” below refers to the finite proxy registry unless it is explicitly qualified as source-level or asymptotic. No endpoint exponent is inferred from the table.

2 The finite source and operator

Let N be even and set

$$\mathcal{I}_N = \{N/2 + 1, \dots, N\}, \quad \mathcal{Q}_Q = \{q : q \text{ prime}, Q < q \leq 2Q\}.$$

For an integer $z \geq 2$, let $b_N^{(z)}$ be the finite shifted-prime comparison from TPC-268:

$$b_N^{(z)}(u) = C_2^{(z)} 1_{2 \nmid u} \prod_{p \leq z} \frac{p}{p-1} \prod_{\substack{p|u \\ p > z}} \frac{p-1}{p-2},$$

where the value is zero when $u + 2$ has a prime factor at most z , and $C_2^{(z)} = \prod_{p>z} (1 - (p-1)^{-2})$. Define

$$\beta_N(t) = \frac{\Lambda(t)}{\log t} - \sum_{\substack{d|t \\ d^{400} \leq N^{133}}} \mu(d), \quad w_N(u) = \Lambda(u+2) - b_N^{(z_N)}(u).$$

The finite cutoff proxy is

$$z_N = \lfloor \log N \rfloor, \quad (z_{64}, z_{96}, z_{128}, z_{192}, z_{256}, z_{384}) = (4, 4, 4, 5, 5, 5).$$

For $s = 1, 2$, use $K_{H,s}(h) = (1 + (h/H)^2)^{-s}$. These are the two finite kernel representatives inherited from the previous audit. If A_s denotes (with the same shell, masks, and deleted diagonal)

$$A_s(u, t) = 1_{u \neq t} \sum_{q \in \mathcal{Q}_Q} q K_{H,s}(u-t) 1_{q \nmid ut} \left(1_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right),$$

then for rational $0 \leq \theta \leq 1$ define

$$A_\theta = (1 - \theta)A_1 + \theta A_2, \quad g_\theta = A_\theta \beta_N.$$

3 Exact profile transfer

Partition $\mathcal{I}_N$ into four equal consecutive blocks of size $B = N/8$ and use the orthogonal contrasts

$$c_0 = (1, 1, -1, -1), \quad c_1 = (1, -1, 0, 0), \quad c_2 = (0, 0, 1, -1).$$

If W_j and G_j are the block sums of w_N and g_θ , put

$$C_\theta = \langle w_N, g_\theta \rangle, \quad C_{3,\theta} = \frac{W_0 G_0}{4B} + \frac{W_1 G_1}{2B} + \frac{W_2 G_2}{2B}, \quad C_{\perp,\theta} = C_\theta - C_{3,\theta}.$$

Let

$$R_\theta^2 = \|(I - P_3)w_N\|^2 \|(I - P_3)g_\theta\|^2, \quad \rho_\theta^2 = \frac{|C_{\perp,\theta}|^2}{R_\theta^2}.$$

Proposition 1 (affine transfer identity). *For every finite row and rational θ , $g_\theta = (1 - \theta)g_1 + \theta g_2$ and $C_{\perp,\theta} = (1 - \theta)C_{\perp,1} + \theta C_{\perp,2}$. The projection identity $C_\theta = C_{3,\theta} + C_{\perp,\theta}$ is exact.*

Proof. All entries of A_1, A_2 are rational on a finite row. Linearity of matrix multiplication and of the orthogonal projection gives the first two identities. The three contrasts are pairwise orthogonal, so the displayed denominators are their squared norms. Subtracting the projected cross-Gram proves the last identity. The denominator R_θ^2 is quadratic in θ , so the proposition deliberately makes no monotonicity assertion about ρ_θ . $\square$

4 Interval certificate

The Euler products are multiplied through $P = 50000$. The positive tail bound

$$\prod_{p>P} \left(1 - \frac{1}{(p-1)^2} \right) \geq 1 - \sum_{p>P} \frac{1}{(p-1)^2} > 1 - \frac{1}{P-1}$$

Table 1: Selected rows. Values are rounded square-root upper endpoints; the decision uses the stored ρ^2 interval.

N	H	Q	z_N	θ	displayed ρ	class
64	15	4	4	0	0.2735358969	obstruct
96	20	5	4	0	0.2996937679	obstruct
128	24	5	4	0	0.1248321291	contract
192	32	6	5	0	0.0072691733	contract
256	38	6	5	0	0.0338322886	contract
384	50	7	5	0	0.0310909465	contract
64	15	4	4	9/10	0.2518350028	obstruct
64	15	4	4	24/25	0.2495257250	contract
96	20	5	4	1/2	0.2985106861	obstruct
128	24	5	4	1/2	0.1421214943	contract
256	38	6	5	1/2	0.0157231342	contract

gives rational endpoints. Logarithms use 100-digit decimal intervals with a 10^{-25} guard, and all endpoints are rounded outward to a 10^{-30} grid. The decision is made before taking a square root:

$$\rho_{\text{hi}}^2 < \frac{1}{16} \Rightarrow \text{contraction}, \quad \rho_{\text{lo}}^2 > \frac{1}{16} \Rightarrow \text{obstruction}.$$

The producer computes the two exact rational operator outputs separately and forms their rational convex combination. The independent checker uses an independent floating-point implementation; it is an audit rather than a substitute for the interval enclosure.

Theorem 1 (finite growing-cutoff/profile transfer). *The certificate contains twelve threshold-separated rows, eight contractions and four obstructions. Its six base rows use $\theta = 0$ and the cutoff $z_N = \lfloor \log N \rfloor$. In the central row $(N, H, Q) = (64, 15, 4)$,*

θ	ρ^2 interval	classification
9/10	[0.0634078324659, 0.0634208686352]	obstruction
24/25	[0.0622500850692, 0.0622630874560]	contraction
1	[0.0614513775060, 0.0614643508731]	contraction

so the convex profile path crosses the quarter threshold while the cutoff, clock, shell, and projection are fixed.

Proof. The finite identities are the proposition. The interval producer enumerates the registered rows and propagates rational endpoints through the projection and quotient. Every contraction upper endpoint is below $1/16$ and every obstruction lower endpoint is above $1/16$. The independent replay reproduces all twelve strict inequalities and the stress audit checks the cutoff registry and the profile flip. These checks establish the stated finite theorem. $\square$

5 Interpretation and limits

The finite proxy does two useful things. It removes the arbitrary fixed $z = 2$ from the experiment and makes the profile dependence explicit through an affine path. It also gives a sharp warning: a favorable endpoint profile does not imply a profile-uniform phase bound, because ρ_θ is a quotient with a changing denominator.

Three promotions are prohibited. The schedule $z_N = \lfloor \log N \rfloor$ is not a proof of the source-level V59 cutoff theorem. The twelve rows do not control R_θ at growing scale and hence pay no fixed-power credit. Finally, the finite profile path is not an arithmetic L^2 estimate and does not close the collective prime-shell reassembly in Gate B.

6 Conclusion

TPC-269 supplies a finite transfer obstruction with a new source-compatible proxy: a growing cutoff and a convex profile family on the same physical operator. The central 9/10 versus 24/25 flip shows that even this canonicalized finite interface is not uniformly inside the quarter sector. The next minimal task is cross-scale normalization of the residual radius, with the profile and cutoff held source-compatible. The actual V59 radius, phase, arithmetic L^2 , strict 1/400 payment, and twin-prime conclusion remain open.

References

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